Rucha Mene

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Summary

Final-year Computer Science student (CGPA 8.4/10) with an AI engineering internship shipping 20+ front-end modules and 150+ Playwright tests, 500+ DSA problems solved, and a custom L4/L7 load balancer in Java. Seeking SDE roles.

Education

Vellore Institute of Technology Oct 2022 – May 2026

B.Tech in Computer Science and Engineering ***(CGPA: 8.4/10)***

Experience

AI Software Engineer Intern Feb 2026 – Aug 2026

Jai AI HCE Secure Pvt Ltd (AI Startup) *Indore, India*

- Delivered **LLM-powered features** – a chat interface and semantic search UI – for a secure enterprise AI assistant, giving employees natural-language access to internal knowledge.
- Engineered **20+ front-end modules** using **React.js, TypeScript, and Electron**, including secure authentication and **REST API integrations** with existing backend services.
- Automated environment setup, process management, and result checks using **Bash and PowerShell** scripts alongside **Playwright**, making test execution repeatable across **VMware-based CentOS, Rocky Linux, and Ubuntu** environments kept current with **OS patches and package updates**.
- Maintained **Docker-based CI/CD pipelines** and authored **150+ Playwright end-to-end tests** to catch regressions before releases.
- Collaborated with backend and product teams to ship features from requirements, tracking **incidents and defects in Jira**.

Projects

Load Balancer | *Java, Docker, Prometheus, Grafana* GitHub

- Architected an **L4/L7 load balancer** in Java (socket programming) across **5+ containerized backend instances** with **4 routing algorithms**: Round Robin, Weighted Round Robin, Least Connections, Consistent Hashing.
- Devised a **Token Bucket rate limiter** with configurable per-client throttling, tested under **1,000+ simulated requests**.
- Benchmarked **throughput, latency, and load-distribution fairness**, exposing **10+ Prometheus metrics** on a Grafana dashboard.

AI Career Coach | *React.js, Shadcn UI, OpenAI API, Supabase, Clerk* GitHub

- Designed an **AI-powered career guidance platform** with **4+ core workflows** (resume feedback, skill-gap analysis, roadmaps, interview prep), using the **OpenAI API** to turn user inputs into personalized recommendations across **5+ scenarios**.
- Implemented secure authentication with **Clerk** and persistent storage with **Supabase**, supporting user profiles, career progress, and recommendation history.
- Created **10+ reusable React components** with **Shadcn UI** for resume analysis, skill tracking, and career planning.

Crypto Tracker | *React.js, Node.js, MongoDB, JavaScript, CoinGecko API* GitHub

- Launched a **real-time cryptocurrency tracking platform** covering **100+ coins** via the CoinGecko API, with **5+ features**: live prices, conversion, trend analysis, historical charts, and forecasting.
- Established **Node.js REST API** workflows with **MongoDB** for structured data management, including asynchronous data fetching and error handling.
- Optimized client-side rendering with **lazy loading and efficient API handling**, reducing unnecessary network requests.

Technical Skills

Languages: C++, Java, Python, JavaScript, TypeScript, SQL, Bash
Frontend & Backend: React.js, Next.js, Electron, Node.js, Express.js, Spring Boot, FastAPI, REST APIs
Databases: PostgreSQL, MySQL, MongoDB, Redis
AI/ML: LLM Integration, RAG, Semantic Search, OpenAI API, Gemini API
DevOps & Testing: Docker, CI/CD, Git, GitHub, Linux, Playwright, Prometheus, Grafana
CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

Achievements

- **Contributor, GirlScript Summer of Code (GSSoC) 2024** – Contributed to open-source projects.
- Solved **500+ Data Structures and Algorithms** problems across multiple competitive coding platforms.
- **Hackathon Winner** – Won a college-level coding hackathon by turning an idea into a working prototype.